

Paloma — Project Overview

B.Sc. Computer Science final project. This document explains what the project is, what was built, and how it was verified. By Yonatan Gabay, yonatang8675@gmail.com.

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1. What this project is

Paloma is a two-stage processing pipeline for **TESS Full Frame Images** (FFIs). TESS is a NASA space telescope that photographs large patches of sky repeatedly, so that brightness changes over time reveal transiting exoplanets, supernovae, and other variable phenomena.

Two problems stand between a raw FFI and a usable light curve:

1. **Scattered light.** Sunlight and earthshine scatter inside the telescope optics and add a large, smoothly varying glow across the frame. It swamps faint sources and drifts between frames.
2. **Crowding.** TESS pixels are 21 arcseconds wide, so many stars fall in one pixel. Measuring one star directly is unreliable; what you actually want is the *change* relative to a stable reference.

Paloma addresses both, in order:

```
raw FFIs  -> Stage 1: Cleaning (dehazer)  -> cleaned FITS
          -> Stage 2: Image Subtraction (OIS)
          -> residual images + source catalog + light curves
```

Stage 1 removes the scattered-light glow. Stage 2 builds a reference frame from the cleaned data, subtracts each frame from it to leave only what changed, then detects sources in those residuals and extracts a light curve for each one.

The engineering contribution of this project is not the two algorithms themselves — they came from existing research code. It is that both were **rewritten from a specification into one tested, configurable, composable pipeline** whose numerical output is verified against the originals.

2. Where the code came from

Upstream	What it does	Origin
<code>tess-scatter-removal</code> (private)	Scattered-light removal via spatio-temporal patch recurrence	Accompanies the M.Sc. thesis <i>Removing Scattered Light in TESS Full Frame Images via Spatio-Temporal Patch Recurrence</i> , Shachar Fridman, Ariel University, 2026
<code>image-subtraction</code>	Optimal Image Subtraction for TESS FFIs, Python + a C differencing binary	Based on the Oelkers et al. implementation

Both upstream READMEs are kept next to this file in `docs/` (`tess-scatter-removal README.md`, `Image Subtraction Readme.md`) as a record of the original interfaces.

The dehazer treats scattered light exactly like **haze in a photograph**. That is the key insight worth understanding, because the whole algorithm follows from it. The standard atmospheric haze model is

$$I(x) = L(x) t(x) + A (1 - t(x))$$

where I is what the detector records, L is the true scene, $t \in (0,1]$ is transmission (how much of the true scene survives), and A is the *airlight* — the intensity of the scattering veil. Recovering L means estimating A and t , then inverting the equation. Techniques from blind image dehazing therefore transfer to astronomical frames.

3. What was done, step by step

The work was not a direct port. Each algorithm was locked down with tests, specified from the existing code, then rewritten from that specification. That order is the point: the tests pin what the original actually does, the reverse PRD says what it is supposed to do, and the rewrite has to satisfy both.

The same loop ran twice — first for the dehazer, then for image subtraction — and only then were the two stages composed into one pipeline.

- Clone the original.** Start from `tess-scatter-removal` (private).
- Write tests against it, including a ground-truth test.** Run the original on real TESS FFIs, capture its output, and treat that as the numerical contract. From this point on, a rewrite is allowed to change structure, not pixels.
- Write a reverse PRD from the code.** Infer the intended behaviour from the implementation rather than from comments or a missing spec.
- Rewrite from the PRD only.** Implement 3D (multi-frame) mode only, keep the tests green, and restructure so the algorithm is readable and configurable. The original 2D (per-frame) path was

dropped.

- 5. **Repeat steps 1–4 for image subtraction**, starting from `image-subtraction`.
- 6. **Compose both algorithms into one chainable pipeline**. Each stage is a strategy behind a registry, so the pipeline calls an interface rather than a concrete implementation. New cleaners or subtractors register; they do not get wired in by hand.
- 7. **Cover the whole pipeline with tests and a `.env` file**. Stage selection, paths, and algorithm knobs all come from environment configuration, and the wiring is tested independently of the expensive science loops.

4. Verification status

Both rewrites are checked against output captured from the original implementations, on the same **5 real TESS FFIs** (sector s0003-2-1, 2048×2048). Reproduce with `pytest tests`.

```
55 passed in 271.67s (0:04:31)
```

Stage 1 — dehazer

`tests/cleaning/test_groundtruth.py` compares every pixel of the cleaned frames to the original dehazer's FITS. Tolerance is `rtol = atol = 1e-9`. At that bound the rewrite is numerically equivalent to the original.

Stage 2 — image subtraction

`tests/image_subtraction/test_groundtruth.py` asserts the **full stage output** — residuals, source catalog, and light curves — in one run. Fixtures live under `tests/data/image_subtraction/groundtruth/`; details are in [that directory's README](#).

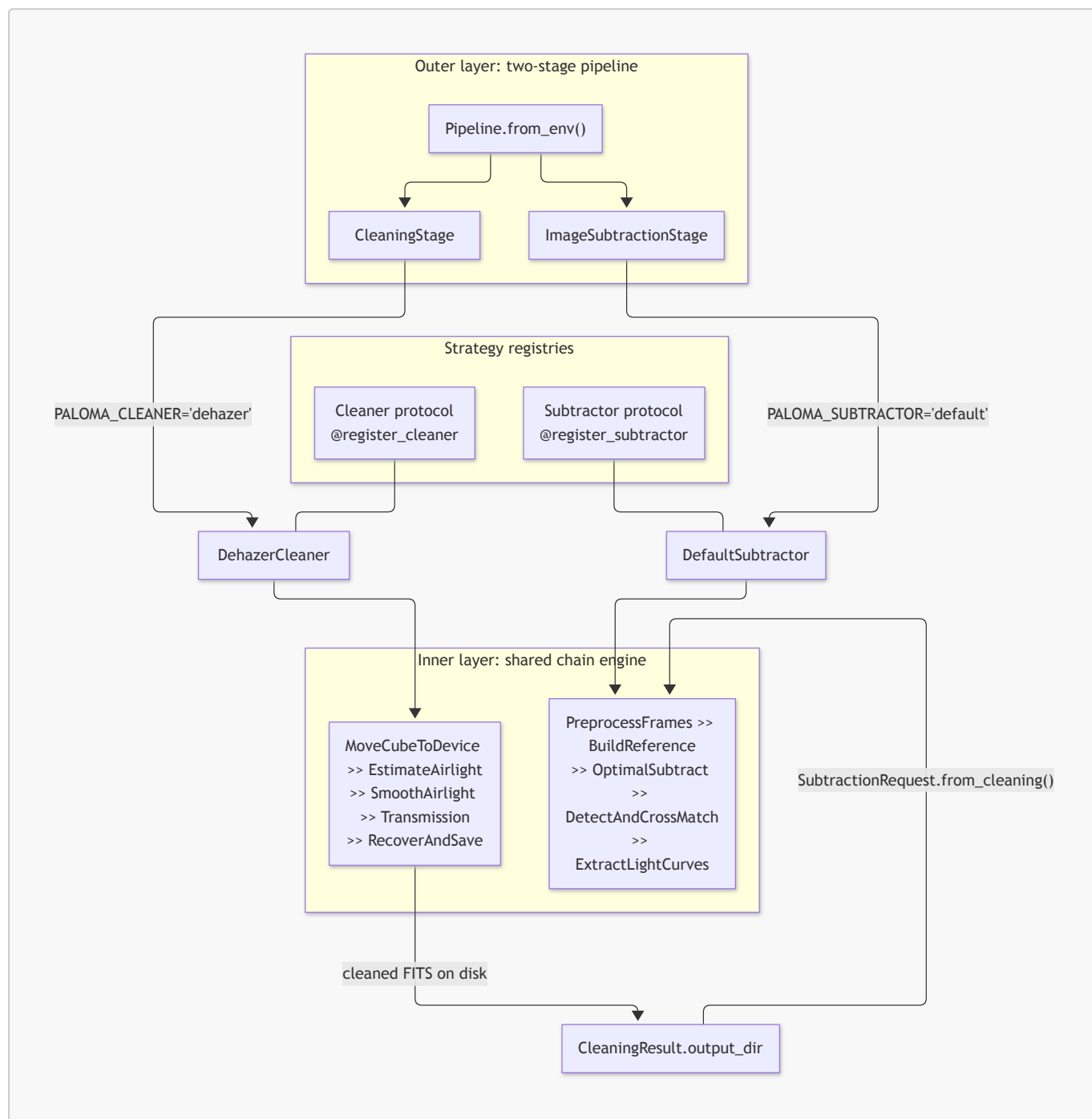
Artifact	What is compared	Contract
Residual FITS (5 frames)	Paloma vs the original C-backend residuals	<code>rtol = 1e-6, atol = 1e-2</code> . Largest observed deviation is <code>7.8e-3</code> , which is half an ULP of float32 at the brightest pixel (<code>~1.4e5</code>). Residuals are stored as float32, so the two backends agree as closely as the file format permits.
<code>found_sources.csv</code> (1922 rows)	Paloma vs the original detection / cross-match (<code>src/core_science.py</code> , commit <code>456b2d2</code>)	Row-for-row numeric match at <code>rtol = atol = 1e-9</code>
<code>lightcurves.fits</code>	Paloma vs the original photometry on those residuals	<code>TIME</code> to <code>1e-9</code> ; <code>FLUX / FLUX_ERR</code> at <code>rtol = 1e-4, atol = 0.5</code> (a 3-pixel aperture on residuals that differ by <code>~1e-2</code> counts)

5. Architecture

Two layers. Both algorithms use them the same way.

Outer layer — the pipeline. Cleaning then image subtraction. Each stage is a plug-in: the pipeline calls an interface, and `.env` chooses which algorithm fills it. Stages pass data through directories, not in-memory arrays.

Inner layer — a shared chain engine. Each algorithm is a list of steps joined with `>>`, all writing into one context object.



The strategy layer

Cleaning and image subtraction are the same plug-in slot, twice. The pipeline never names a concrete algorithm. It asks the registry for whoever `.env` selected.

	Cleaning	Image subtraction
Interface	Cleaner	Subtractor
Env key	PALOMA_CLEANER	PALOMA_SUBTRACTOR
Built-in	"dehazer"	"default"
Knobs	PALOMA_DEHAZE_*	PALOMA_SUBTRACTION_*

Both interfaces are a name plus one method. Return `None` if nothing usable was produced; that stops the pipeline.

```
class Cleaner(Protocol):
    name: str
    def clean(self, request: CleaningRequest) -> Optional[CleaningResult]:
    ...

class Subtractor(Protocol):
    name: str
    def subtract(self, request: SubtractionRequest) ->
Optional[SubtractionResult]: ...
```

To add an algorithm: subclass `BaseCleaner` or `BaseSubtractor`, decorate it with `@register_cleaner` or `@register_subtractor`, and point the matching env key at its `name`. Do not edit the pipeline. `paloma/__init__.py` imports both packages so the built-ins register on import.

The files are `paloma/core/cleaner.py` and `paloma/core/subtractor.py`.

The chain engine

`paloma/core/chain.py` is shared by both algorithms. A `Stage` has a `name`, an optional `requires` tuple, and an `apply(ctx)` method. `>>` composes stages into a `Chain`, plain callables are wrapped automatically, and `requires` is enforced at runtime against `ctx.run_history`, so an out-of-order chain fails loudly instead of producing garbage.

Each registered strategy builds one chain and runs it against its own context:

Strategy	Context	Chain
DehazerCleaner	WorkflowContext	MoveCubeToDevice >> EstimateAirlight >> SmoothAirlight >> Transmission >> RecoverAndSave
DefaultSubtractor	SubtractionWorkflowContext	PreprocessFrames >> BuildReference >> OptimalSubtract >>

Strategy	Context	Chain
		DetectAndCrossMatch >> ExtractLightCurves

A trap: `None` means different things in the two layers. In the outer pipeline, a strategy returning `None` halts everything. In the chain engine, a step returning `None` just means "I mutated the context in place" and execution continues. Both are intentional; the asymmetry is easy to misread.

The handoff

The two strategies do not share memory. Cleaning writes cleaned FITS; subtraction reads that directory. That keeps peak memory bounded and makes intermediate results inspectable.

`SubtractionRequest.from_cleaning()` in `paloma/core/types.py` takes `CleaningResult.output_dir` as its input directory and overrides two defaults, because cleaned frames are written as a bare `PrimaryHDU`:

```
defaults = {
    "apply_data_quality_filter": False, # cleaned frames have no DQUALITY
    "fits_extension": 0,               # science data is in HDU 0, not 1
}
```

6. Algorithm 1 — the dehazer

Entry point: `dehaze()` in `dehazer/runner.py`. **Chain:** `MoveCubeToDevice >> EstimateAirlight >> SmoothAirlight >> Transmission >> RecoverAndSave`.

Scattered light is treated as haze. The same small star-field patch appears many times in a frame, under different amounts of glow. Matching those patches estimates the glow (airlight A) and how much of the true scene survives (transmission t). No training data — the frames supply their own statistics.

$$I = L t + A(1 - t)$$

The chain, in order:

- 1. Load.** Read FITS, crop calibration edges, normalize the whole batch to $[0, 1]$ with one shared min/max (not per frame — otherwise later time-smoothing is meaningless). Filename order is the time axis.
- 2. Estimate airlight.** Keep structured 9×9 patches, match similar ones, and solve for A from each pair (p_1, p_2) (with mean-centred versions t_1, t_2). Average the estimates, weighting pairs that differ most in haze.

$$A_{12} = \langle t_2 - t_1, p_1 \odot t_2 - p_2 \odot t_1 \rangle / \|t_2 - t_1\|^2$$

- 3. Smooth airlight.** Scattered light changes smoothly in time, so $A(t)$ is Gaussian-smoothed across frames. Single-frame estimates are noisy; this step is doing real work.

4. Transmission. Build a per-pixel map from the haze model, then guided-filter it so edges follow the stars, not the noise. Smooth that map in time too.

$$t \approx 1 - I / A$$

5. Recover. Invert the model, undo the normalization, write `dehazed_*.fits` in original units.

$$L = (I - A) / t + A$$

Knobs are `PALOMA_DEHAZE_*`.

7. Algorithm 2 — optimal image subtraction

Entry point: `subtract()` in `image_subtraction/pipeline.py`. **Chain:** `PreprocessFrames >> BuildReference >> OptimalSubtract >> DetectAndCrossMatch >> ExtractLightCurves`.

You cannot subtract two frames directly. The PSF jitters between exposures, so a plain difference rings around every star. **Optimal Image Subtraction** fits a convolution kernel that makes the reference look like the science frame, then subtracts. What remains actually changed.

The chain, in order:

1. Preprocess. Optional TESS quality filter, cutout, align every frame onto the first frame's WCS. Output: `01_preprocessed/`.

2. Build the reference. Median-combine frames into block masters, then into `03_ref_data/master_final.fits`. Median rejects cosmic rays; the stack is deeper than any single frame. Detect stars on it for the kernel fit.

3. Subtract. Remove background, fit a small kernel on those stars, convolve the reference, subtract. A positive residual is extra flux in the science frame. Two passes by default: rebuild the reference and subtract again. Output: `04_residual_img/`. Python and C backends are numerically equivalent; C is faster if `build/a.out` is present.

$$residual = science - (kernel * reference)$$

4. Detect and cross-match. Find both brightenings and fadings on each residual (`DA0StarFinder` also looks for negative peaks). Drop frames with far too many detections, then match sources across frames into `05_sources/found_sources.csv`.

5. Light curves. Aperture photometry on each residual. One FITS per source in `06_lightcurves/`, with `TIME`, `FLUX`, `FLUX_ERR`.

Knobs are `PALOMA_SUBTRACTION_*`.

8. Repository layout

```
paloma/
__main__.py      # python -m paloma (full pipeline)
pipeline.py      # Pipeline, CleaningStage, ImageSubtractionStage
```

```
config.py  env.py      # .env loading and typed accessors
core/
  chain.py      # the >> engine: Stage, Chain, as_stage, timed
  cleaner.py    # Cleaner protocol + registry
  subtractor.py # Subtractor protocol + registry
  stage.py      # PipelineStage ABC (outer layer)
  types.py      # Cleaning/Subtraction Request + Result
dataclasses
  log.py        # stdout progress helpers, warning filters
cleaning/
  dehazer_cleaner.py # adapter: Cleaner -> dehaze()
  dehazer/
    runner.py    # dehaze(): batching, GPU resolution,
orchestration
  cli.py        # dehaze | simulate | validate
  config.py     # DehazeConfig
  core/         # backend, patches, airlight, guided_filter,
recover
  io/           # fits_io, layout
  workflow/     # context + the 5 chain stages
  evaluation/   # synthetic data + MSE/PSNR/SSIM validation
image_subtraction/
  pipeline.py   # subtract() + DefaultSubtractor adapter
  cli.py  config.py
  core/         # preprocess, masters, background, ois, ois_c,
                # kernel_stars, sources, photometry
  io/layout.py  # RunLayout, the 01..06 output directories
  workflow/     # context + the 5 chain stages
tests/          # see section 11
docs/           # this file, plus the two upstream READMEs
.env.example    # the authoritative list of every PALOMA_* key
pyproject.toml
```

Output directories for a subtraction run, created by `RunLayout.create()`:

Directory	Contents
01_preprocessed/	Aligned, cut-out frames; plus <code>rejectedFiles.txt</code>
02_masters_frames/	<code>master_000.fits</code> , ... — block medians
03_ref_data/	<code>master_final.fits</code> , its star list, <code>.flux</code>
04_residual_img/	<code>subtracted_*.fits</code> (float32)
05_sources/	<code>found_sources.csv</code>
06_lightcurves/	One FITS per source with a <code>LIGHTCURVE</code> table

9. Configuration

Every tunable lives in `.env`. `.env.example` is the authoritative catalog and is kept in sync with `ENV_KEYS` in `paloma/env.py` by `tests/test_env.py` — that test fails if you add a key to one and not

the other. Start with `cp .env.example .env`.

Coercion rules: an empty value means "use the code default"; booleans accept `true/false/1/0/yes/no/on/off`; pairs are `x,y`; integers are parsed with base 0, so `0b110010111101` and `0xFF` both work; relative paths resolve against the repository root.

Selection

Key	Default	Meaning
<code>PALOMA_CLEANER</code>	<code>dehazer</code>	Which registered cleaner to use
<code>PALOMA_SUBTRACTOR</code>	<code>default</code>	Which registered subtractor to use
<code>PALOMA_INPUT_DIR</code> , <code>PALOMA_OUTPUT_DIR</code>	—	Pipeline I/O roots
<code>PALOMA_CLEANED_DIR</code> , <code>PALOMA_SUBTRACTED_DIR</code>	<code>{output}/cleaned</code> , <code>{output}/subtracted</code>	Per-stage overrides

Dehazer — `PALOMA_DEHAZE_*`

Key suffix	Default	Effect
<code>PATCH_SIZE</code>	9	Patch side length in pixels
<code>VARIANCE_THRESHOLD</code>	<code>5e-5</code>	Minimum patch variance to keep
<code>NN_DIST_THRESHOLD</code>	0.3	Descriptor distance cutoff for pairing
<code>NUM_ITERATIONS</code>	10	Airlight reweighting rounds
<code>SIGMA_TEMPORAL</code>	1.5	Gaussian sigma for temporal smoothing
<code>T_MIN_CLIP</code>	0.01	Transmission floor
<code>GUIDED_FILTER_RADIUS</code>	60	Guided filter window radius
<code>GUIDED_FILTER_EPS</code>	0.001	Guided filter regularization
<code>NUM_FRAMES</code>	10	Maximum frames to load
<code>BATCH_SIZE</code>	empty	Frames per batch (empty = all at once)
<code>CROP_BOTTOM</code> , <code>CROP_SIDES</code>	30, 44	Calibration-edge crop
<code>MAX_PATCHES</code>	empty	Cap on patches per frame
<code>FITS_EXTENSION</code>	1	Science HDU index
<code>USE_GPU</code>	<code>false</code>	Opt in to CuPy if a free CUDA device exists

Changing any of the first eight changes numeric output and will break the `1e-9` ground-truth test. That is intended — the test pins the algorithm.

Image subtraction — `PALOMA_SUBTRACTION_*`

Key suffix	Default	Effect
<code>KERNEL</code>	2	Kernel half-width; footprint is $(2k+1)^2$
<code>STAMP</code>	3	Fitting stamp half-width
<code>ORDER</code>	0	Spatial polynomial degree of the kernel
<code>NR_STARS</code>	500	Kernel stars to fit against
<code>BLKNUM</code>	50	Frames per block master
<code>NUM_ITERATIONS</code>	2	Subtract/rebuild passes
<code>THRESHOLD_SOURCE_DETECTION</code>	50.0	Detection threshold in sigma
<code>FWHM</code>	1.5	<code>DA0StarFinder</code> FWHM in pixels
<code>MATCH_RADIUS</code>	3	Cross-match radius in pixels
<code>APERTURE_RAD</code>	3	Photometry aperture radius
<code>EDGE_CUTOFF</code>	3	Border pixels excluded from detection
<code>SOURCE_FILTER_THRESHOLD</code>	1.5	Frame-rejection sigma on detection counts
<code>CUTOUT_CENTER, CUTOUT_SIZE</code>	1068, 1024, 2048, 2048	Cutout geometry
<code>CRPIX1</code>	1001.0	Reference <code>CRPIX1</code> used when aligning and cutting out
<code>NUM_FRAMES</code>	empty	Frame cap (empty = every FITS in the input directory)
<code>FITS_EXTENSION</code>	1	Science HDU index (the handoff overrides this to 0)
<code>APPLY_DATA_QUALITY_FILTER</code>	true in code, false in <code>.env.example</code>	Reject frames by TESS DQUALITY bits
<code>DATA_QUALITY_MASK</code>	0b110010111101	TESS quality bits to reject
<code>USE_C_BACKEND</code>	true	Prefer <code>build/a.out</code> , else fall back to Python
<code>CODE_DIR</code>	None in code, <code>build/code_dir</code> in <code>.env.example</code>	Scratch directory for the C backend
<code>RANDOM_SEED</code>	0	Seeds kernel-star selection

10. How to run it

```
python -m venv .venv
source .venv/bin/activate
pip install -e ".[dev]"
cp .env.example .env
```

Python 3.10 or newer. Note that `pyproject.toml` pulls `FITS_tools` straight from GitHub, so the install needs network access.

Full pipeline:

```
python -m paloma
python -m paloma --input-dir path/to/fits --output-dir results
```

Or from Python, which is also how you compose the stages yourself:

```
from paloma import Pipeline
result = Pipeline.from_env().run()
```

```
from paloma import (
    CleaningStage, CleaningConfig, CleaningRequest,
    ImageSubtractionStage, ImageSubtractionConfig,
)

cleaned = CleaningStage.from_config(CleaningConfig.from_env()).run(
    CleaningRequest(input_dir="...", output_dir="results/cleaned")
)
sub = ImageSubtractionStage.from_config(ImageSubtractionConfig.from_env())
result = sub.run_after_cleaning(cleaned, "results/subtracted")
```

Single stages:

```
python -m paloma.cleaning.dehazer dehaze --input-dir ... --output-dir ...
python -m paloma.cleaning.dehazer simulate --clean-fits ... --type cloud
python -m paloma.cleaning.dehazer validate --sim-dir ... --output-dir ...
python -m paloma.image_subtraction subtract --input-dir ... --output-dir ...
...
```

Optional C backend for OIS:

```
gcc oisdifference.c -lcfitsio -lm -o build/a.out
```

`oisdifference.c` is **not** in this repository; get it from the upstream `image-subtraction` repo. It is faster than the Python path but no longer more accurate.

A copy-paste first run — clone, smoke-test the five bundled FFIs, then point `--input-dir` at your data — is the [appendix](#).

11. Testing strategy

```
pytest tests          # 55 tests, ~4.5 minutes (ground-truth tests
                        dominate)
pytest tests -q --deselect tests/cleaning/test_groundtruth.py \
                --deselect tests/image_subtraction/test_groundtruth.py  #
~1s
```

Three layers, covering both stages the same way.

Ground-truth. Pins the science: a live run on 5 real TESS FFIs is compared to fixtures under `tests/data/`, so a refactor cannot silently change pixels, catalogs, or light curves. Missing fixtures skip the test. Image-subtraction `raw/` is a symlink onto the cleaning frames, so the FITS are not stored twice.

Test	What it checks
<code>tests/cleaning/test_groundtruth.py</code>	Every pixel of the cleaned FITS (<code>rtol = atol = 1e-9</code>)
<code>tests/image_subtraction/test_groundtruth.py</code>	Residuals, <code>found_sources.csv</code> , and light curves

Tolerances for the subtraction artifacts are in [section 4](#).

Wiring. The pipeline is only names and directories — these tests check that stages still plug together, config still loads, and a `None` result still stops the run, without waiting for the science loops. Those loops are mocked.

- `tests/cleaning/test_integration.py`
- `tests/image_subtraction/test_integration.py`
- `tests/test_pipeline.py`
- `tests/test_env.py`

Unit. A ground-truth failure tells you *that* something broke, not *where*. These cover one function at a time, on tiny arrays, so a kernel or timestamp bug shows up in seconds.

- `tests/core/test_chain.py` — the `>>` engine
- `tests/image_subtraction/test_ois.py` — kernel fit, including a flux-rescale case whose residual should be `~1e-13`

- `tests/image_subtraction/test_photometry.py`
- `tests/image_subtraction/test_sources.py`

Appendix — Minimal example

The smallest useful run is the five bundled TESS FFIs (sector s0003-2-1) under `tests/data/cleaning/groundtruth/raw/`. That is the same set used in [section 4](#). Once it finishes, a production job is the same command with a different `--input-dir`.

Prefer a **CPU** node. Image subtraction never uses a GPU. The dehazer can, but only if CuPy is installed, `PALOMA_DEHAZE_USE_GPU=true`, and the node is exclusive enough to pass the free-VRAM check. CPU allocations are simpler and cover the whole pipeline. Do not compile `build/a.out` unless you already have `cfitsio`; without it Paloma falls back to the Python OIS, which matches the C residuals to float32 precision.

1. One-time setup

Install needs outbound network: `pyproject.toml` pulls `FITS_tools` from GitHub. Put the venv in `$HOME` — scratch is often `noexec` or cannot host venv symlinks.

```
module load python/3.11          # site module, Python ≥ 3.10
git clone https://github.com/Or-Deri/TESS-Data-pipeline.git paloma
cd paloma
python -m venv "$HOME/paloma-venv"
source "$HOME/paloma-venv/bin/activate"
pip install -U pip
pip install -e .
cp .env.example .env
```

2. Smoke test

```
source "$HOME/paloma-venv/bin/activate"
cd /path/to/paloma
python -m paloma \
  --input-dir tests/data/cleaning/groundtruth/raw \
  --output-dir "$SCRATCH/paloma-smoke"
```

A few minutes. Success prints the cleaned-frame count, then the residual, catalog, and light-curve paths. Look for:

```
$SCRATCH/paloma-smoke/cleaned/dehazed_*.fits
$SCRATCH/paloma-smoke/subtracted/04_residual_img/
$SCRATCH/paloma-smoke/subtracted/05_sources/found_sources.csv
$SCRATCH/paloma-smoke/subtracted/06_lightcurves/
```

If import fails on `libGL.so.1`, `pip install opencv-python-headless`.

3. Run on your own FFIs

Put the science `*.fits` in one directory; alphabetical order is the time axis. In `.env`, set `PALOMA_DEHAZE_NUM_FRAMES` to at least that count — the default cap is 10. Leave `PALOMA_DEHAZE_USE_GPU=false`. `--input-dir / --output-dir` override `.env`; everything else stays there.

```
#!/bin/bash
#SBATCH --job-name=paloma
#SBATCH --cpus-per-task=8
#SBATCH --mem=32G
#SBATCH --time=04:00:00
#SBATCH --output=paloma-%j.out
# Add --partition / --account if your site requires them.

module load python/3.11
source "$HOME/paloma-venv/bin/activate"
cd /path/to/paloma

python -m paloma \
  --input-dir "$SCRATCH/tess/s0003-2-1" \
  --output-dir "$SCRATCH/paloma-out"
```

```
sbatch paloma.slurm
```

After the venv exists, a later dataset is that one `sbatch`. Write outputs to scratch, not `$HOME`.